LAB ASSIGNMENT -2

Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.

Data Members(properties):

Name
Roll No
Degree
Hostel
Current CGPA

Users > Shagun Garg > oops.cpp > ...

```cpp
#include <iostream>
#include <string>
using namespace std;

struct Student {
    string Name;
    int RollNo;
    string Degree;
    string Hostel;
    float CurrentCGPA;

    void addDetails() {
        cout << "Enter Name: ";
        cin >> Name;

        cout << "Enter Roll No: ";
        cin >> RollNo;

        cout << "Enter Degree: ";
        cin >> Degree;

        cout << "Enter Hostel: ";
        cin >> Hostel;

        cout << "Enter Current CGPA: ";
        cin >> CurrentCGPA;
    }

    void updateCGPA(float newCGPA) {
        CurrentCGPA = newCGPA;
        cout << "CGPA updated successfully." << endl;
    }
```

```
void updateHostel(string newHostel) {
    Hostel = newHostel;
    cout << "Hostel updated successfully." << endl;
}

void displayDetails() {
    cout << "\nStudent Details" << endl;
    cout << "Name: " << Name << endl;
    cout << "Roll No: " << RollNo << endl;
    cout << "Degree: " << Degree << endl;
    cout << "Hostel: " << Hostel << endl;
    cout << "CGPA: " << CurrentCGPA << endl;
}


main() {
Student s[2];

for(int i = 0; i < 2; i++) {
    cout << "\nEnter details for Student " << i + 1 << endl;
    s[i].addDetails();
}

for(int i = 0; i < 2; i++) {
    s[i].displayDetails();
}
```

```
    float newCGPA;
    for(int i = 0; i < 2; i++) {
        cout << "\nEnter new CGPA for Student " << i + 1 << ": ";
        cin >> newCGPA;
        s[i].updateCGPA(newCGPA);
    }

    string newHostel;
    for(int i = 0; i < 2; i++) {
        cout << "Enter new Hostel for Student " << i + 1 << ": ";
        cin >> newHostel;
        s[i].updateHostel(newHostel);
    }

    for(int i = 0; i < 2; i++) {
        s[i].displayDetails();
    }

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\

Enter details for Student 1
Enter Name: Shagun
Enter Roll No: 1024150104
Enter Degree: ENC
Enter Hostel: Q
Enter Current CGPA: 9.56

Enter details for Student 2
Enter Name: Vanshika
Enter Roll No: 1024030123
Enter Degree: COE
Enter Hostel: E
Enter Current CGPA: 9.68

Student Details
Name: Shagun
Roll No: 1024150104
Degree: ENC
Hostel: Q
CGPA: 9.56

Student Details
Name: Vanshika
Roll No: 1024030123
Degree: COE
Hostel: E
CGPA: 9.68
```

```
Enter new CGPA for Student 1: 9.99
CGPA updated successfully.

Enter new CGPA for Student 2: 9.73
CGPA updated successfully.
Enter new Hostel for Student 1: N
Hostel updated successfully.
Enter new Hostel for Student 2: N
Hostel updated successfully.

Student Details
Name: Shagun
Roll No: 1024150104
Degree: ENC
Hostel: N
CGPA: 9.99

Student Details
Name: Vanshika
Roll No: 1024030123
Degree: COE
Hostel: N
CGPA: 9.73
PS C:\Users\Shagun Garg>
```

2. Differentiate between private and public access/scope. Perform the question no. 1 with class instead of structure with having the data members private and some member functions in private scope and some in public scope.

Users > Shagun Garg > oops.cpp > Student > displayDetails()

```cpp
#include <iostream>
#include <string>
using namespace std;

class Student {
    private:
    string Name;
    int RollNo;
    string Degree;
    string Hostel;
    float CurrentCGPA;

    public:
    void addDetails() {
        cout << "Enter Name: ";
        cin >> Name;

        cout << "Enter Roll No: ";
        cin >> RollNo;

        cout << "Enter Degree: ";
        cin >> Degree;

        cout << "Enter Hostel: ";
        cin >> Hostel;

        cout << "Enter Current CGPA: ";
        cin >> CurrentCGPA;
    }

    void updateCGPA(float newCGPA) {
        CurrentCGPA = newCGPA;
        cout << "CGPA updated successfully." << endl;
    }

    void updateHostel(string newHostel) {
        Hostel = newHostel;
        cout << "Hostel updated successfully." << endl;
    }

    void displayDetails() {
        cout << "\nStudent Details" << endl;
        cout << "Name: " << Name << endl;
        cout << "Roll No: " << RollNo << endl;
        cout << "Degree: " << Degree << endl;
        cout << "Hostel: " << Hostel << endl;
        cout << "CGPA: " << CurrentCGPA << endl;
    }
};
```

```cpp
int main() {
    Student s[2];

    for(int i = 0; i < 2; i++) {
        cout << "\nEnter details for Student " << i + 1 << endl;
        s[i].addDetails();
    }

    for(int i = 0; i < 2; i++) {
        s[i].displayDetails();
    }

    float newCGPA;
    for(int i = 0; i < 2; i++) {
        cout << "\nEnter new CGPA for Student " << i + 1 << ": ";
        cin >> newCGPA;
        s[i].updateCGPA(newCGPA);
    }

    string newHostel;
    for(int i = 0; i < 2; i++) {
        cout << "Enter new Hostel for Student " << i + 1 << ": ";
        cin >> newHostel;
        s[i].updateHostel(newHostel);
    }

    for(int i = 0; i < 2; i++) {
        s[i].displayDetails();
    }

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\

Enter details for Student 1
Enter Name: Shagun
Enter Roll No: 1024150104
Enter Degree: ENC
Enter Hostel: S
Enter Current CGPA: 9.00

Enter details for Student 2
Enter Name: Riya
Enter Roll No: 1024303000
Enter Degree: ECE
Enter Hostel: Y
Enter Current CGPA: 8.77

Student Details
Name: Shagun
Roll No: 1024150104
Degree: ENC
Hostel: S
CGPA: 9

Student Details
Name: Riya
Roll No: 1024303000
Degree: ECE
Hostel: Y
CGPA: 8.77
```

```
Enter new CGPA for Student 1: 9.55
CGPA updated successfully.

Enter new CGPA for Student 2: 8.23
CGPA updated successfully.
Enter new Hostel for Student 1: A
Hostel updated successfully.
Enter new Hostel for Student 2: S
Hostel updated successfully.

Student Details
Name: Shagun
Roll No: 1024150104
Degree: ENC
Hostel: A
CGPA: 9.55

Student Details
Name: Riya
Roll No: 1024303000
Degree: ECE
Hostel: S
CGPA: 8.23
PS C:\Users\Shagun Garg> ^C
PS C:\Users\Shagun Garg>
```

3. Create a code snippet that illustrates the following: Calling of private member functions inside public member function

```
Users > Shagun Garg > Untitled-1.cpp > main()
#include <iostream>
using namespace std;

class Demo {
private:
    void privateFunction() {
        cout << "This is a private member function." << endl;
    }

public:
    void publicFunction() {
        cout << "Public function calling private function..." << endl;
        privateFunction();
    }
};

int main() {
    Demo obj;
    obj.publicFunction();
    }
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun G
Public function calling private function...
This is a private member function.
PS C:\Users\Shagun Garg>
```

4. Define a class Rectangle with variables width and height of integer typwe along with following:

(a) void getdata() to initialize object values
(b) void calculatearea() to calculate and display the area

Users > Shagun Garg > oops.cpp > rectangle > getdata()

```cpp
#include <iostream>
using namespace std;

class rectangle {
private:
    int length;
    int breadth;

public:
    void getdata() {
        cout << "Enter length: ";
        cin >> length;
        cout << "Enter breadth: ";
        cin >> breadth;
    }

    void displaydata() {
        cout << "Length: " << length << endl;
        cout << "Breadth: " << breadth << endl;
    }

    void area() {
        cout << "Area of rectangle: " << length * breadth << endl;
    }
};

int main() {
    rectangle r;
    r.getdata();
    r.displaydata();
    r.area();
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\"
Enter length: 4
Enter breadth: 8
Length: 4
Breadth: 8
Area of rectangle: 32
PS C:\Users\Shagun Garg>
```

5. Define a class Complex with variables real and imaginary along with

   following: (a)void setComplex (float, float) to initialize object values.
   (b) void displayComplex() to show the complex number
   (c) Pass and return objects to calculate sum of two complex numbers. Display the sumr.

```
Users > Shagun Garg > oops.cpp > ...
#include <iostream>
using namespace std;

class Complex {
private:
    float real;
    float imaginary;

public:
    void setComplex(float r, float i) {
        real = r;
        imaginary = i;
    }

    void displayComplex() {
        if (imaginary >= 0)
            cout << real << " + " << imaginary << "i" << endl;
        else
            cout << real << " - " << -imaginary << "i" << endl;
    }

    Complex addComplex(Complex c) {
        Complex temp;
        temp.real = real + c.real;
        temp.imaginary = imaginary + c.imaginary;
        return temp;
    }
};

int main() {
    Complex c1, c2, c3;
    c1.setComplex(3.5, 2.5);
    c2.setComplex(1.5, -4.0);
    c3 = c1.addComplex(c2);
    cout << "First Complex Number: ";
    c1.displayComplex();
    cout << "Second Complex Number: ";
    c2.displayComplex();
    cout<<endl;
    cout << "Sum of Complex Numbers: ";
    c3.displayComplex();
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\"
First Complex Number: 3.5 + 2.5i
Second Complex Number: 1.5 - 4i

Sum of Complex Numbers: 5 - 1.5i
PS C:\Users\Shagun Garg>
```

6. Implement *scope resolution operator* : : for the following uses:
    i. Class functions defined outside the class
    ii. Access a global variable with same name as a local variable
    iii. Access a static variables

Use inbuilt libraries (cin cout with scope resolution operator)

Users > Shagun Garg > oops.cpp > getdata()

```cpp
#include <iostream>
using namespace std;
class Rectangle{
    int length;
    int bredth;
    void displaydata(){
        cout<<"Length of rectangle: "<<length<<endl;
        cout<<"Bredth of rectangle: "<<bredth<<endl;
    }
    public:
    void getdata();
    void Area();
};
int main() {
    Rectangle r;
    r.getdata();
    r.Area();
    return 0;
}
void Rectangle::getdata(){
        cout<<"Enter length of rectangle: ";
        cin>>length;
        cout<<"Enter bredth of rectangle: ";
        cin>>bredth;
}
void Rectangle::Area(){
        displaydata();
        int area=length*bredth;
        cout<<"Area of rectangle: "<<area<<endl;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Use
Enter length of rectangle: 5
Enter bredth of rectangle: 9
Length of rectangle: 5
Bredth of rectangle: 9
Area of rectangle: 45
PS C:\Users\Shagun Garg>
```

7. Create a code to implement the *namespace* and use similar variables and functions defined in different code sections.

Users > Shagun Garg > oops.cpp > { } SectionB

```cpp
#include <iostream>
using namespace std;
namespace SectionA {
    int value = 10;

    void show() {
        cout << "SectionA value: " << value << endl;
    }
}
namespace SectionB {
    int value = 20;

    void show() {
        cout << "SectionB value: " << value << endl;
    }
}
int main() {
    cout << "Accessing variables from namespaces:" << endl;
    cout << "SectionA value = " << SectionA::value << endl;
    cout << "SectionB value = " << SectionB::value << endl;

    cout << "\nCalling functions from namespaces:" << endl;
    SectionA::show();
    SectionB::show();

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users
Accessing variables from namespaces:
SectionA value = 10
SectionB value = 20

Calling functions from namespaces:
SectionA value: 10
SectionB value: 20
PS C:\Users\Shagun Garg>
```